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Stochastic Optimization of Spatial Throughput in Wireless Networks under Determinantal Scheduling

Project Context and Motivation

Domain: Wireless communication and network optimization

Real-world Problem:

- In wireless networks, we want two things:
- Good coverage (reliable connection)
- Fast speed (high data transfer)
- But it’s hard to get both at the same time.

Why is this uncertain?

- Signals fade randomly
- Devices interfere with each other
- Too many devices = overload

Too few devices = slow speed

Project Objective and Scope Main Goal:

- Find the best number of devices (node density) that gives the highest total speed (throughput).
- What are we estimating?

  Throughput = Number of active devices × Chance of successful transmission

  We want to find the optimal density that gives maximum throughput.
- Scope:

  Compare two scheduling methods:

  Aloha (random access)

  DPP (smart access with repulsion)

Project System Overview Step-by-step system:

- Generate random device positions
- Apply scheduling method (DPP or Aloha)
- Calculate SINR (signal quality)
- Compute coverage probability
- Multiply by density → throughput
- Compare results
- Data Flow:
  Devices → Scheduler → SINR → Coverage → Throughput → Comparison

Sources of Uncertainty in the Project

- Measurement Noise:

  Signal strength changes randomly (Rayleigh fading)

- Environmental Variability:

  Device positions and interference patterns change

- Modelling Uncertainty:

  We assume certain formulas that may not match real world perfectly

Key Random Variables and Their Roles

- Node Density (λ): How many devices try to transmit
- Fading (F): Random signal strength
- Interference (I): Noise from other transmitters
- Coverage Probability (P): Chance that SINR > threshold
- Throughput (T): $\lambda \times P$ (total data speed)

Probabilistic Models and Assumptions

Models Used:

- Poisson Point Process for node locations
- Exponential distribution for fading

Assumptions:

- Fading variables are independent
- Noise power is constant

Justification:

- These models are standard and match real wireless behavior.

Probabilistic Reasoning and Inference Logic

Uncertainty Propagation:

Random locations → random interference → random SINR → random success.

Step-by-Step Logic:

- Nodes are placed randomly
- Scheduling selects active nodes
- Interference is calculated
- SINR is evaluated
- Success probability is obtained
- Throughput is computed

Operationalizing Probabilistic Reasoning in the Project

How we implement it:

- Use existing Python/MATLAB code
- Change fairness metric to throughput
- Run simulations for different densities

Evaluation Metrics:

- Coverage probability
- Total throughput
- Comparison between Aloha and DPP

Simplifying Assumptions:

- Ignore edge effects
- Assume fading is independent and identical

Current Limitations and Conceptual Gaps

Limitations:

- Only single-hop communication is considered
- Assumes ideal independence in fading

Conceptual Gaps:

- Optimal density is not analytically derived
- No real-world experimental validation yet

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Planned Refinements and Role Coordination

Next Steps:

- Implement throughput optimization
- Plot density vs. throughput curves
- Compare peak throughput for DPP and Aloha

Modeling Tasks Ahead:

- Study how interference changes with density
- Analyze the peak point of throughput curves

Summary of Group-Level Understanding

- Wireless networks are uncertain by nature
- Probability helps model interference and fading
- Throughput depends on both density and success probability
- Determinantal scheduling improves spatial reuse
- Optimal density gives maximum network performance